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Eddies: Dynamical Action Tokens with Certified Composition

Supplementary Material

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Appendix A. The Execution Layer

This appendix proves that every command step of Streamer is well defined and that the supervisor finishes a task or refuses it before the robot moves. Neither fact uses the identified models (the tracking certificates do: Appendix E). Equation numbers without a letter, Theorem 1 and Algorithm 1 refer to the main paper. The projection that turns a proposal into the next command must respect both limits of (1).

Proposition A.1 (The projection is well posed) *Let Eddie i run at step k, and let the current command lie in its trust ball, $\|e_k - \bar{e}_i\|_M \le \delta_i$. Let D_k be the set of commands that respect both limits at the next step, $D_k = \{e : \|e - \bar{e}_i\|_M \le \delta_i,\ \|e - e_k\|_M \le \rho_i\}$. Then D_k is nonempty, closed and convex, and so for every proposal p exactly one command $e_{k+1} \in D_k$ is closest to p in $\|\cdot\|_M$. The same holds at a hand-over to Eddie i', with its anchor $\bar{e}_{i'}$, trust radius $\delta_{i'}$ and slew bound $\rho_{i'}$, whenever the current command is close enough to the new anchor, $\|e_k - \bar{e}_{i'}\|_M \le \delta_{i'} + \rho_{i'}$.*

Proof. D_k contains e_k (the trust condition is the assumption; the slew condition holds with distance zero) and is the intersection of two closed balls of $\|\cdot\|_M$, hence compact and convex. The squared distance $\|e - p\|_M^2$ is continuous and, since $M \succ 0$, strictly convex in e, so it has exactly one minimizer on D_k. At a hand-over only nonemptiness needs a new argument, since e_k may lie outside the new trust ball. Let $d = \|e_k - \bar{e}_{i'}\|_M$. If $d = 0$, e_k is the new anchor and lies in both balls. Otherwise the point $e_k + \min(d, \rho_{i'})(\bar{e}_{i'} - e_k)/d$ has distance $\min(d, \rho_{i'}) \le \rho_{i'}$ to e_k and $\max(d - \rho_{i'}, 0) \le \delta_{i'}$ to $\bar{e}_{i'}$, so it lies in both balls. The implementation also intersects D_k with the box of valid normalized commands, which is convex and contains e_k and the anchors, hence the point above, which lies on the segment between them, so the argument is unchanged. ■

If instead $d_k = \|e_k - \bar{e}_{i'}\|_M > \delta_{i'} + \rho_{i'}$, no command lies in both the new trust ball and the slew ball, and Streamer bridges: it takes the point of the new trust ball closest to e_k and projects it onto the slew ball alone. Each step brings the command at least $\rho_{i'}$ closer to the anchor, so Proposition A.1 applies after at most $\lceil (d_k - \delta_{i'} - \rho_{i'})/\rho_{i'} \rceil$ steps. Meanwhile the slew bound still holds, and it is the only one of the two limits that the tube (4) needs.

The second fact rests on the symbolic contracts of the primitives (Section 4.3 of the main paper): the labels that must hold before the primitive's Eddie starts and the outcomes it may produce. A product state $x = (\ell, y)$ pairs the current label cell ℓ, which includes counters such as a retry budget, with the state y of a finite automaton for the goal φ (De Giacomo and Vardi, 2013); the set X of

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product states is finite. For an Eddie i allowed at x, $\mathsf{Post}(x, i) \subseteq X$ is the nonempty set of product states in which its contract allows it to end. The contracts are sound if Eddie i, started anywhere in the cell of x, ends in $\mathsf{Post}(x, i)$ within H steps whenever the gate lets it finish. Let $\mathcal{V}_0$ hold the product states whose automaton state is accepting, and let

$$\mathcal{V}_{n+1} = \mathcal{V}_n \cup \{x : \ \mathsf{Post}(x, i) \subseteq \mathcal{V}_n \text{ for some Eddie } i \text{ allowed at } x\}. \tag{A.1}$$

The rank of x is $r(x) = \min\{n : \ x \in \mathcal{V}_n\}$.

Theorem A.2 (Finish or refuse) *Let the contracts be sound, and let x_0 be the product state before the robot moves. The rank of x_0 then decides between two cases. (a) Suppose that $r(x_0) < \infty$. Consider the supervisor that, at every hand-over, measures the product state x and runs an Eddie allowed at x whose outcomes, the states in $\mathsf{Post}(x, i)$, all have rank below $r(x)$. Suppose also that the gate lets every Eddie finish. Then this supervisor reaches $\mathcal{V}_0$ within $r(x_0)$ Eddies, and so within $r(x_0)H$ steps, and the label trace satisfies φ. This holds for every sequence of outcomes that the contracts allow. (b) Suppose that $r(x_0) = \infty$. Then the contracts allow outcomes that keep any supervisor out of $\mathcal{V}_0$ forever. In this case Streamer refuses φ before the robot moves.*

Proof. The sets $\mathcal{V}_n$ grow inside the finite set X, so they stop growing at some $\mathcal{V}_*$, and a product state has finite rank exactly when it lies in $\mathcal{V}_*$. (a) Let the measured product state x have rank $n \geq 1$. By (A.1) some allowed Eddie i has $\mathsf{Post}(x, i) \subseteq \mathcal{V}_{n-1}$, whose states all have rank below n, so the supervisor always has an Eddie to run, and soundness places the product state measured afterwards among that Eddie's outcomes. So the rank drops by at least one per Eddie and reaches zero within $r(x_0)$ Eddies of at most H steps each. A product state of rank zero lies in $\mathcal{V}_0$, so its automaton state is accepting, and by then the automaton has read the whole label trace, so the trace satisfies φ. (b) Let $x \notin \mathcal{V}_*$ and let i be any Eddie allowed at x. Then $\mathsf{Post}(x, i) \not\subseteq \mathcal{V}_*$, since otherwise x would belong to the next set, which is again $\mathcal{V}_*$. So the contract of Eddie i allows an outcome outside $\mathcal{V}_*$, to which the same argument applies, and so on forever. Since $r(x_0) = \infty$, x_0 is outside $\mathcal{V}_*$, and since $\mathcal{V}_0 \subseteq \mathcal{V}_*$, no supervisor reaches $\mathcal{V}_0$. Streamer refuses in advance because it computes $\mathcal{V}_*$ before motion. This is the standard attractor (strong-preimage) construction for reachability games, applied to the product with the goal automaton as in strong planning for LTL_f goals (Camacho and McIlraith, 2019). ■

On the Franka (Section 5 of the main paper), the product game over the Franka's Eddies recovers the four hand-written plans from the formulas of Table C.2, solves five new goals, each of which succeeds in 99–100% of 100 episodes, and refuses three impossible goals before motion, because their initial rank is infinite ($r(x_0) = \infty$, Theorem A.2b).

Appendix B. Identification and Gains

Identification

Commands and states live in a normalized chart: position and rotation vector in $[-1, 1]$, hand shape and force in $[0, 1]$. The command e_k is the full command that the controller receives (Table B.1): hand position and rotation vector, plus the gripper aperture and a force limit on the Franka, the closure of the Dex3 hand on the G1, and the power, pinch and point synergies on the LEAP and Aero hands. The state z_k is the measured hand pose in the same chart, recorded before e_k is applied,

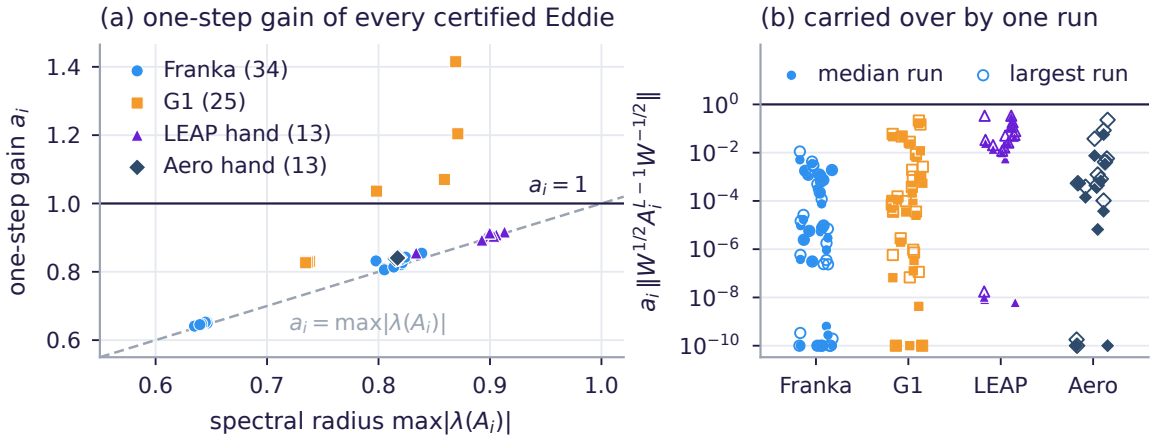


Figure B.1: (a) One-step gain $a_i = \|W^{1/2} A_i W^{-1/2}\|$ of every certified Eddie of the four robots against the spectral radius of A_i; the dashed diagonal (equality) is a lower bound on a_i. (b) The factor $a_i \|W^{1/2} A_i^{L-1} W^{-1/2}\|$ with which one run of L steps passes its starting bound to the next Eddie: median and largest factor over the Eddie's training runs, values below 10^{-10} drawn at 10^{-10}. 81 of the 85 Eddies of the ten tasks have $a_i < 1$ (Theorem 1 of the main paper).

without the coordinates whose pooled standard deviation on the training data is at most 0.01 (the rotation about y on the LEAP and Aero hands); hand shape and force thus enter only through B_i. S picks the position rows of z and scales them to metres; M and W are one hand-set diagonal per robot, M on all of e and W its pose part, neither fitted to data.

Table B.1: Identification per robot: dimensions m of e and n of z; weights of M in the order of e (pose; hand shape, force), of which W is the pose part; S in metres per normalized unit on x, y, z; and the reference Eddie, toward whose model (B.1) shrinks all of the robot's models.

Robot	m	n	weights of M	S [m]	reference Eddie
Franka	8	6	1, 1, 1, 0.3, 0.3, 0.3; 0.15, 0.05	0.35, 0.5, 0.35	releasing the box
G1 + Dex3	7	6	1, 1, 1, 0.3, 0.3, 0.3; 0.05	0.4, 0.4, 0.4	grasping the peg
UR5e + LEAP / Aero	9	5	1, 1, 1, 0.3, 0.3, 0.3; 0.05, 0.05, 0.05	0.35, 0.5, 0.35	releasing the peg

Each task has 200 training, 200 calibration and 200 test episodes, and the model of Eddie i uses the training episodes alone. With regressors $\phi_k = (z_k, e_k, 1)$ and parameters $\Theta_i = [A_i \; B_i \; d_i]$, it predicts z_{k+1} as $\Theta_i \phi_k$, fitted over the n_i transitions of the robot's training episodes in which Eddie i runs at step k, including the last step of each run, with a penalty toward a reference model Θ_{ref}:

$$\Theta_i = \arg\min_{\Theta} \sum_k \|z_{k+1} - \Theta\phi_k\|^2 + \lambda\, n_i\, \|\Theta - \Theta_{\text{ref}}\|_F^2. \tag{B.1}$$

Θ_{ref} is the model of the robot's reference Eddie (Table B.1), the Eddie with the smallest training tube among those with at least 5,000 training transitions and a stable fit of their own; the penalty makes the fit unique when some command channels are constant within an Eddie. The weight $\lambda \in \{0, 0.03, 0.3, 3\}$, where 0 stands for a tiny ridge toward zero, is chosen per Eddie as the value that gives a stable fit with the smallest ultimate tube on the training data. Shrinkage matters most

for Eddies with few transitions: the Eddie that searches for the hole has 148 transitions, and its tube is 193 cm with its own fit and 2.7 cm with shrinkage.

The residuals w_k of the selected model set κ_i, the 90th percentile over the training episodes of the largest $\|W^{1/2}w_k\|$ that Eddie i produces in each; q then comes from the disjoint calibration episodes. A model is certified only if its spectral radius is below 0.999 and it has at least 50 transitions. On the ten tasks this certifies 34 of the Franka's 35 Eddies (the one left out, backing off from the hole with the peg, has a stable fit but only 16 transitions), 25 G1 Eddies and 13 on each of the LEAP and Aero hands. Every Eddie of every nominal plan is certified, and the largest certified spectral radius is 0.913. The self-calibrating gate (Algorithm 1 of the main paper) refits a model after each run of its Eddie by (B.1), with the Eddie's prior in place of Θ_{ref} and, in place of λn_i, a fixed weight of 200, 50 or 10 pseudo-transitions for an Eddie, primitive or global prior (one model of all the robot's labelled training steps). The refit uses all of that Eddie's transitions in the task's episodes so far (up to a stop), except the last transition of each run, and is kept only if its spectral radius is below 0.995; as the model is frozen during a run, it is used from the Eddie's next run, which may come later in the same episode (staged insertion runs five Eddies twice). κ_i keeps the prior's value until the Eddie has run 5 times and is then the 90th percentile of the Eddie's own largest residuals per run. For a new task, the priors come from the other tasks only.

One-step gains at hand-overs

The hand-over bound (7) holds for any one-step gain $a_i = \|W^{1/2}A_iW^{-1/2}\|$, and even when $a_i \geq 1$ the chained bound does not compound: within an Eddie the tube (4) contains the powers A_i^j, which decay whenever A_i is stable, whereas a_i enters once per hand-over. A run of L steps thus passes its starting bound on with the run factor of Fig. B.1(b), which is small in practice, so the chained bound is set by $J_{ii'}$, ϵ_i and the sums in (4). On the Franka, the LEAP hand and the Aero hand every Eddie has $a_i < 1$, with gains of 0.64–0.85, 0.86–0.92 and 0.84, each at most 0.034 above the spectral radius of its model (equal for a normal matrix, so these models are close to normal in the metric W). On the G1, four Eddies have $a_i > 1$, all reaching moves of the drawer task; the largest gain is 1.41, for reaching to grasp the caddy, with spectral radius 0.87. Even so, the factor over a whole run stays below one on every robot: at most 0.011 on the Franka (lifting the peg, $L = 23$), 0.21 on the G1, 0.34 on the LEAP hand (releasing the peg, about 12 steps) and 0.23 on the Aero hand, and usually orders of magnitude smaller.

Appendix C. Benchmark Details

The ten tasks (Fig. 6 of the main paper) are built from the eleven primitives of Table C.1. Each robot implements its own vocabulary once and shares it across its tasks: the Franka implements all eleven (its four tasks use all but turn), the G1 all but turn, and the LEAP and Aero hands reach, grasp, push, release, turn and retreat. A task is an Eddie stream: its robot's primitives in order, each bound to one of the task's frames (Table C.2).

The tasks run in MuJoCo (Todorov et al., 2012) at 500 Hz, with elliptic friction cones, joint damping, friction loss and actuator force limits, and with masses and friction coefficients from measurements or the literature. Every robot is driven at 50 Hz: damped least-squares inverse kinematics on the hand point turns each command into targets for joint position servos, and the command also sets the hand shape (gripper aperture on the Franka, Dex3 closure on the G1, finger synergies on the LEAP and Aero hands). The Franka's insertion tasks add a force-limiting admittance loop that

Table C.1: The eleven primitives from which each robot's vocabulary is drawn, and the taxonomies that list them: Bøgh et al. (2012), MAPLE (Nasiriany et al., 2022), RAPS (Dalal et al., 2021) and FMB (Luo et al., 2025). A primitive bound to a frame is a skill; a skill with its identified model and envelope is an Eddie. Proposition E.4 splits the error budget when a primitive is added; in Appendix D, the scale κ_i is part of each Eddie and the Eddies of a task share one calibrated coverage level (Lemma E.1).

Primitive	Behaviour	Listed by
reach	move the hand to a frame	MAPLE (reach), RAPS (go to pose)
grasp	close the hand; success is measured pad contact	Bøgh et al. (pick up), MAPLE, RAPS, FMB
lift	lift the held object	RAPS
transport	carry the held object to a frame, faces aligned	cf. Bøgh et al. (move to), FMB (move to board)
place	lower the held object onto a support, force-limited	Bøgh et al., RAPS (drop), FMB (place on fixture)
release	open the hand in place	Bøgh et al., MAPLE, RAPS
insert	force-limited compliant insertion, optional spiral search	cf. Bøgh et al. (align), FMB
push	force-limited push with closed fingers	Bøgh et al. (press), MAPLE, RAPS
pull	pull a grasped articulation along its identified line	RAPS
turn	rotate the grasp about an axis	Bøgh et al., RAPS (twist), FMB (rotate)
retreat	open and move clear of the object	(post-release clearance)

reads a filtered, noisy wrist force sensor; the G1 keeps its balance by leg inverse kinematics and centre-of-mass regulation; the LEAP hand has joint position servos and the Aero hand pull-only tendon servos.

Table C.2: The ten tasks. The robots come from MuJoCo Menagerie (Zakka et al., 2022). Eddies: number of Eddies in each nominal plan. Each episode is judged against the listed LTL_f formula on the complete measured label trace; *bad* collects collisions, excessive contact forces, drops and unrecoverable states.

Robot	Task	Eddies	LTL_f specification	Horizon [s]
Franka	Staged insertion	16	`G!bad & F(grasped & F(staged & F(grasped & F(inserted & F(seated & released)))))`	90
	Peg insertion	8	`G!bad & F(grasped & F(inserted & released))`	60
	Cube stack	8	`G!bad & F(grasped & F(stacked & released))`	45
	Drawer + box	14	`G!bad & F(opened & F(grasped & F(placed & released)))`	80
G1 + Dex3	Peg insertion	8	`G!bad & F(grasped & F(inserted & released))`	60
	Drawer + caddy	16	`G!bad & F(opened & F(grasped & F(placed & released)))`	75
UR5e + LEAP / Aero	Wheel	6	`G!bad & F(grasped_rim & F(turned & F released))`	45
	Peg tightening	7	`G!bad & F(seated & F released)`	90

Staged insertion, peg insertion and cube stacking are also certified in Isaac Sim 5 (NVIDIA, 2026a,b) with the same task, method and controller code (Fig. C.1): the compiled MuJoCo scene of every task and seed is converted to USD with Isaac Sim's MJCF importer, Isaac Sim integrates all dynamics with a 2 ms step, and after each physics step the states of the articulations and free objects are written back into the task's MuJoCo data, so MuJoCo's forward kinematics still supplies



Figure C.1: Peg insertion in Isaac Sim, from the same task, method and controller code as the MuJoCo runs.

the sites that the task logic reads. Servos become implicit Isaac Sim drives with the MuJoCo gains and force limits, and colliders keep their MuJoCo friction.

In Table 2 of the main paper, a borrowed certificate keeps the source's models, normalizers and coverage level; a re-identified certificate is fitted and calibrated on the target, for the hands on the target's own training and calibration episodes, for Isaac Sim on the MuJoCo data plus 10 Isaac Sim episodes per task, calibrated on 50 more. Cube stacking in Table 3 uses 50 episodes for the fit and 50 for calibration. Finished counts the runs that the fixed-calibration gate lets finish at $\mu = 15$ cm.

The real-robot experiment is cube stacking on a Franka Panda, with the eight Eddies of Table C.2. The arm runs under the 1 kHz compliant controller of CRISP (San José Pro et al., 2026), which tracks the 50 Hz Eddie commands. The cube poses come from a static RealSense camera, calibrated hand-to-eye, in the robot base frame from two sources: boxes in the RGB image, and heights above the table from the aligned depth (Fig. 8 of the main paper). The re-identified certificate is refit on the MuJoCo data plus 50 robot episodes and calibrated on 50 more. At nominal speed there are 50 test runs, half of them perturbed by pushing a cube with a stick during the approach. Both gates run online on the same measurements: the fixed-calibration gate acts on the arm, and the stops of the self-calibrating gate are only logged. The fast runs are 50 at each of 2, 3 and 4 times the nominal speed, executed with no gate acting and both gates replayed on their logs, as in simulation. The two certificates run on the same 50 seeds. The safe set is the set of cube poses for which the reach Eddie's check (5) of the main paper passes; a cube pushed out of it makes the Eddie refuse its next command, and the arm holds its last command, a certified stop, until the cube is back.

Appendix D. Result Tables

Tables D.1 and D.2 give the per-task numbers at 2 and 5 cm behind Figs. 5 and 7 of the main paper, with the gates replayed on the logged test episodes.

Pooled over the ten tasks at 5 cm, 890 of 2,000 runs finish with the calibration fixed and the plan checked before motion, 1,968 with only the next command checked by (5), and 1,955 with the self-calibrating gate (Table D.1: the last two differ on the LEAP wheel and by one run on each G1 task). At 3 cm, 395 runs would leave μ; the self-calibrating gate stops 387 of them first, and 1,328 runs finish.

The drawer coverage of 0.835 in Table D.3 (167 of 200 episodes) is below the target of 0.9 but within what exchangeability allows: with 200 calibration and 200 test episodes, the coverage conditional on the calibration set is $\mathrm{Beta}(181, 20)$, the number of covered test episodes is beta-binomial with a standard deviation of about 3% in coverage, and a coverage of 0.835 or less has probability 0.026 for one pre-specified task and 0.10 for the lowest of four. One model per primitive, shared by all four tasks, is also valid, at the price of a chained bound about twice as large (the "One

Table D.1: Eddie certificates on 200 test episodes per task, each from its own calibration (Section 5.1 of the main paper). Eddies: Eddies of the nominal plan. Viol.: episodes that leave a tube. Tube: median and 90th percentile of the tube radius. Finished: runs the gate lets finish at $\mu = 5$ cm; the fixed gate holds the calibration fixed, checks every planned Eddie's tube before motion and each hand-over from the measured error, and stops at the first residual outside the envelope; the one-step gate also holds the calibration fixed but checks only the next command by (5) of the main paper; the self-calibrating gate (Algorithm 1 of the main paper) adds the refits and the updates of q. Caught, at $\mu = 2$ cm: of the runs that would leave μ, those the self-calibrating gate stops first. No finished run leaves μ. Bold: best value.

					finished at 5 cm			caught
Robot	Task	Eddies	viol.	tube [cm]	fixed	one-step	self-calibrating	at 2 cm
Franka	Staged insertion	16	1	2.9 / 3.6	188	**199**	**199**	193 / 200
	Peg insertion	8	1	3.1 / 3.5	184	**199**	**199**	194 / 200
	Cube stack	8	0	3.4 / 4.3	180	**191**	**191**	175 / 200
	Drawer + box	14	0	3.0 / 4.2	0	**200**	**200**	200 / 200
G1 + Dex3	Peg insertion	8	0	4.8 / 6.6	0	**198**	197	22 / 26
	Drawer + caddy	16	2	4.8 / 7.9	0	**198**	197	200 / 200
LEAP hand	Wheel	6	1	7.4 / 11.9	0	**183**	172	196 / 200
	Peg tightening	7	0	3.9 / 4.6	174	**200**	**200**	200 / 200
Aero hand	Wheel	6	0	6.3 / 6.5	0	**200**	**200**	200 / 200
	Peg tightening	7	0	2.9 / 4.1	164	**200**	**200**	1 / 1
All (85 of them certified)		96	5		890	**1968**	1955	1581 / 1627

Table D.2: Each Franka task held out of training and run through the self-calibrating gate on its first 100 test episodes (Fig. 7 of the main paper; Table D.1 uses all 200), started with no calibration episode of the task or with the task's own calibration as the prior. The conformal certificate finishes no run without calibration episodes and needs about 55 of them to finish 83–93 of 100. Caught: runs stopped before they leave μ, of those that would (none at 5 cm). False stops: runs stopped although they stay within μ. No finished run leaves μ. Bold: best value.

		new task (no calibration)			own calibration as prior		
Held-out task	μ [cm]	caught	false stops	finished	caught	false stops	finished
Staged insertion	2	**99 / 100**	**0**	**0**	**99 / 100**	**0**	**0**
	5	–	2	98	–	**1**	**99**
Peg insertion	2	37 / 100	**0**	**0**	**99 / 100**	**0**	**0**
	5	–	**1**	**99**	–	**1**	**99**
Cube stack	2	17 / 100	**0**	**0**	**91 / 100**	**0**	**0**
	5	–	**4**	**96**	–	**4**	**96**
Drawer + box	2	**100 / 100**	**0**	**0**	**100 / 100**	**0**	**0**
	5	–	5	95	–	**0**	**100**

Table D.3: Franka certificates with one model per Eddie (left) or one model per primitive shared by all four tasks (right); MuJoCo, 200 held-out episodes per task, $\alpha = 0.1$ (Section 5.1 of the main paper). Viol.: episodes that leave the per-Eddie tube or the tube chained from the episode start (Theorem 1(a) of the main paper). Coverage: share of test episodes within the envelope (target 0.9). Bound: median chained bound over the whole episode. Tightness: median over episodes of the largest ratio of measured error to bound over the episode's Eddies. Bold: best value.

		one model per Eddie				one model per primitive		
Task	Eddies	viol. (tube/chain)	coverage	bound [cm]	tightness	viol. (chain)	coverage	bound [cm]
Staged insertion	16	1/1	0.940	**5.2**	0.86	1	0.900	10.8
Peg insertion	8	1/1	0.925	**4.5**	0.86	1	0.910	10.3
Cube stack	8	0/0	0.900	**6.9**	0.86	0	0.900	11.1
Drawer + box	14	0/0	0.835	**8.6**	0.90	0	0.900	15.7

model per primitive" row of Table 1 of the main paper, Franka column); the calibration cannot be shared: one coverage level calibrated over the four Franka tasks pooled covers the mixture, not each task; with one model per primitive the drawer's coverage falls to 0.68, and with one model per Eddie the stack's and the drawer's to 0.87 and 0.83 (per-task calibration, Table D.3, gives 0.90 on every task).

Equilibrium and command

The certificate bounds $\eta_k = z_k - z^\star_{i(k)}(e_k)$, the error to the equilibrium at which the model of the running Eddie settles under e_k, not the error to the command. On the test episodes (seeds 2000–2199), at every executed step of every certified Eddie, we measure the offset $\|S(z^\star_i(e_k) - e_{\mathrm{pose},k})\|$ between the equilibrium and the commanded position e_{pose}. Its median, 95th percentile and maximum are 0.20, 1.53 and 1.92 cm on the Franka, 0.66, 2.98 and 3.24 cm on the G1, 0.59, 1.99 and 2.43 cm on the LEAP hand, and 0.23, 1.82 and 2.44 cm on the Aero hand. The largest offsets occur on the Franka in free-space transports (up to 1.84 cm at the 95th percentile), where shrinkage toward the reference model adds a nearly constant offset, and on the other robots in constrained contact, pulling the drawer and turning the wheel. The measured error to the command, $\|S(z_k - e_{\mathrm{pose},k})\|$, exceeds the ultimate tube of the running Eddie on 0.09% of the Franka steps, 0.88% of the G1 steps, none of the LEAP hand steps and 0.0001% of the Aero hand steps. Almost all Franka exceedances come from searching for the hole, a recovery Eddie outside the nominal plans with only 148 training transitions, whose equilibrium lies 16.4 cm from the command with its own fit and 0.39 cm with shrinkage (hence its tube falls from 193 to 2.7 cm). The offset also turns the certificate into a bound on the error to the command: with O_i the largest offset over the training and calibration commands of Eddie i (0.10–1.91 cm on the Franka), the triangle inequality gives $\|S(z_k - e_{\mathrm{pose},k})\| \le \|S\eta_k\| + O_i$ for commands in that range, so adding O_i to each tube bounds the error to the command; on the test episodes, the offset is at most 1.92 cm.

A learned Eddie proposer

A learned model can replace the supervisor: at every hand-over (episode start, end of an Eddie, re-plan) it predicts the next Eddie, or that the task is done; inside an Eddie nothing changes. It is an MLP with two hidden layers of 64 units whose inputs are the measured hand pose, aperture and force, the task's object positions, the LTL_f propositions, and the previous Eddie with its outcome,

but not the product state, trained by behaviour cloning on the supervisor's choices in the nominal training episodes and 200 perturbed episodes, which contain the recoveries. On held-out hand-overs it predicts the supervisor's choice in 91.7% of cases on stacking, 99.9% on peg insertion and 99.97% on staged insertion, and all recovery hand-overs but one. Guarded by the contracts, a proposal runs only if its contract lowers the rank (Theorem A.2), and the guarded proposer reproduces the supervisor's episodes on all 100 test seeds of every task, nominal and perturbed; unguarded, it still succeeds in 99–100% of the nominal episodes, but under perturbation it issues Eddies again where no contract can win, so the guard is what keeps the finish-or-refuse property. The tracking guarantee does not depend on the proposer: every run that would leave $\mu = 5$ cm is stopped or refused first, also when a behaviour-cloned policy proposes the commands inside the stacking Eddies (it succeeds in 16 of 100 runs, 27 runs would leave μ, and the fixed-calibration gate catches all 27).

Token baselines

Table 1 of the main paper compares Eddies with other ways of cutting a task into tokens, all fitted per robot on the commands at the labelled steps of its training episodes, each dimension scaled so that its 1st and 99th percentiles map to ± 1. BeT clusters single-step commands with k-means and labels each step by its nearest centre (Shafiullah et al., 2022), with as many tokens as the robot has Eddies: 35 on the Franka, 13 on each hand and 25 on the G1. FAST + k-means works on chunks of 1 s (50 steps) (Pertsch et al., 2025): an orthonormal DCT of each dimension of a chunk, coefficients multiplied by ten and rounded, flattened frequency-first and clustered with k-means into the same number of tokens. VQ-BeT trains a residual VQ-VAE with two codebooks of 16 codes on 5-step chunks and keeps the first code (Lee et al., 2024). For both chunked tokenizers, the steps of a chunk share its token. Each token is certified like an Eddie, with an affine model shrunk toward the reference model, a normalizer κ_i, the task's conformal quantile and a tube centred on the model's equilibrium; since the order of the tokens is known only at run time, the plan checked before motion is the set of all tokens the tokenizer emits on the task's training and calibration episodes, and a hand-over is a change of token. Of the remaining rows, one global model gives all steps one token, and one model per primitive pools the bindings of a primitive. The per-step bound sets ϵ_i to the per-step conformal quantile of Eddie i at level α/K, with K the length of the longest calibration episode; this needs at least $K/\alpha - 1$ calibration steps per Eddie, 4,769–22,499 per task on the Franka, which some Eddies of every plan lack, except the Aero wheel plan, which the per-step bound certifies at 10 cm (192 runs). The reference-model row gives every Eddie the robot's reference model (Table B.1) with the Eddie's own κ_i; on the Franka its median one-step prediction error is 8.9–24.8 mm, against 0.15–0.9 mm with each Eddie's own model, and its tube is 18.4, 17.6, 10.5 and 60.1 cm on the Franka, the LEAP hand, the Aero hand and the G1, against 4.3, 11.9, 6.5 and 7.9 cm, 1.5–7.6 times wider.

The Franka uses its four tasks with test seeds 2000–2099; the hands and the G1 use their two tasks with the splits of Table D.1 (training seeds 1000–1199, calibration seeds 1200–1399, test seeds 2000–2199). Each robot thus has 400 test runs (gates replayed on the logged test episodes), and the 95% bootstrap intervals over these runs, stratified by task, are about ± 10 runs per cell: Eddies finish significantly more runs than every other row on the Franka at 5 and 7.5 cm, and on the other robots no other tokens finish a run at the smaller tolerance, although on the G1 the one-reference-model ablation ties with Eddies at 7.5 cm. The tube p90 depends only on the certificate, so its intervals over test runs are negligible. Three further results are not in Table 1. At 10 cm, Eddies finish 359, 174

and 340 runs on the Franka, the LEAP hand and the Aero hand, and the best other row 353 on the Franka (one global model, and also one model per primitive), 178 on the LEAP hand (VQ-BeT; one reference model reaches 181) and 171 on the Aero hand (one reference model), so at 10 cm these rows are level with Eddies on the Franka and the LEAP hand. BeT with about twice as many tokens (64 on the Franka, 26 on the hands, 50 on the G1) comes level with Eddies on the Franka, with a tube of 4.8 cm and 267 and 358 runs at 5 and 7.5 cm at twice the hand-overs per episode (21 against 10.5), but on the other robots it still certifies no plan at the two smaller tolerances, with tubes of 17.8–40.4 cm. On a second Franka split (training seeds 1200–1399, calibration seeds 1000–1199), Eddies finish 269, 362 and 362 runs at 5, 7.5 and 10 cm with a 4.4 cm tube, BeT with 64 tokens 177, 267 and 360 runs with a 5.9 cm tube, and one global model 0, 261 and 352 runs.

Table D.4 compares two ways of certifying a hand-over at tolerance μ with the calibration held fixed, both asking whether the state handed to Eddie i' lies within the starting bound from which its tube (4) stays within μ. Funnel containment (Majumdar and Tedrake, 2017) tests the whole terminal ball of Eddie i, of its ultimate radius around $z_i^\star(e_k)$ and shifted by the jump $J_{ii'}$; the recursion (7) tests the bound actually reached, $a_i b_k + \epsilon_i + J_{ii'}$, carried from the measurement at the start of the episode, and so certifies more at tight tolerances (at 5 cm, 99 of 100 peg-insertion episodes against 1, and 93% of the staged-insertion hand-overs against 79%); the two agree once μ is loose enough for every terminal ball to fit.

Table D.4: Hand-overs and whole episodes certified at tolerance μ, by funnel containment and by the recursion of Theorem 1(a) of the main paper (Franka, 100 test episodes per task, calibration held fixed; Section 5.1 of the main paper). Each cell gives containment / recursion in %. An episode is certified when all its hand-overs are (contribution (ii) of the main paper).

	Staged insertion		Peg insertion		Cube stack		Drawer + box	
μ [cm]	hand-overs	episodes	hand-overs	episodes	hand-overs	episodes	hand-overs	episodes
3	13 / 20	0 / 0	14 / 14	0 / 0	29 / 29	0 / 0	10 / 23	0 / 0
4	47 / 70	0 / 0	43 / 57	0 / 0	34 / 45	0 / 0	23 / 54	0 / 0
5	79 / **93**	0 / 0	86 / **100**	1 / **99**	69 / **86**	0 / 0	69 / **77**	0 / 0
7.5	100 / 100	99 / 99	100 / 100	99 / 99	96 / **100**	75 / **100**	92 / 92	0 / 0
10	100 / 100	99 / 99	100 / 100	99 / 99	100 / 100	100 / 100	100 / 100	100 / 100

In Table D.5, at nominal speed, when no execution is refused, the residual check stops 8–10% of the executions, about α, since with the calibration fixed it fires only on the complement of the event of Lemma E.1. With envelopes calibrated at nominal speed, the gate refuses or stops every execution from $1.5\times$ on, and at 3–4$\times$ it refuses most plans before the robot moves. With envelopes recalibrated at the commanded speed it still completes some executions, 18% and 41% at $1.5\times$ with $\mu = 10$ and 15 cm and 20% at $2\times$ with $\mu = 15$ cm, none of which leaves μ. In Table D.6, pulling the drawer has a 3.0 cm tube on the Franka and 8.3 cm on the G1, whereas reaching above the peg and reaching to the peg stay at 2.9–4.9 cm on all four robots.

Dynamics of the identified Eddies

Here we compare the identified models themselves, using each Eddie's own least-squares fit without the shrinkage of Appendix B, through a descriptor of quantities independent of a robot's coordinate scaling (the time constants of the five slowest modes, the steady-state gains of the position axes and how fast they are reached, and the coupling of the position gains, each scaled by its interquartile

Table D.5: Certify or stop under faster commands (Franka, four tasks; Section 5.1 of the main paper), in % of executions. Leave μ: executions that would leave the tolerance without the gate, computed on the 400 executions calibrated at nominal speed (the other 200 are within 3 points). The other columns give what the gate does with the calibration held fixed, refusing the execution before the start, stopping it at a hand-over or by the residual check, or letting it complete, with envelopes calibrated at nominal speed (400 executions per row) or at the commanded speed on 50 separate episodes per task (200 test executions per row). No completed execution leaves μ.

			calibrated at nominal speed			calibrated at the commanded speed		
speed	μ [cm]	leave μ	refused	stopped	compl.	refused	stopped	compl.
1×	5	0	25	6	69	25	4	70
	10	0	0	10	90	0	8	92
	15	0	0	10	90	0	8	92
1.5×	5	35	100	0	0	100	0	0
	10	2	0	100	0	75	7	18
	15	0	0	100	0	50	9	41
2×	5	97	100	0	0	100	0	0
	10	24	0	100	0	100	0	0
	15	2	0	100	0	75	4	20
3×	5	100	100	0	0	100	0	0
	10	68	75	25	0	100	0	0
	15	22	0	100	0	100	0	0
4×	5	100	100	0	0	100	0	0
	10	95	100	0	0	100	0	0
	15	58	0	100	0	100	0	0

Table D.6: Ultimate tube radius [cm] of the Eddies that run on more than one robot (Section 5.3 of the main paper), each with the calibration q of its own task. A range covers several tasks of one robot; a dash marks a robot without that Eddie.

Eddie	Franka	G1	LEAP	Aero	Eddie	Franka	G1	Eddie	LEAP	Aero
reach peg (above)	3.5–3.6	4.6	4.5	4.1	reach handle (pre)	5.2	4.8	reach rim (pre)	5.6	6.3
reach peg (grasp)	2.9	4.9	3.7	4.5	reach handle	3.2	4.4	reach rim	5.0	4.7
grasp peg	1.8	2.1	2.9	2.7	grasp handle	2.1	7.9	grasp rim	6.2	3.7
lift peg	2.7	4.5	–	–	pull drawer	3.0	8.3	turn wheel	11.9	6.5
transport (above hole)	3.3–3.6	4.8	–	–	release handle	3.1	6.7	release rim	7.4	3.8
insert (hole)	3.1–3.3	6.6	–	–	retreat handle	2.9	4.8	retreat rim	3.9	5.5
release peg	1.9–2.0	2.9	2.8	2.2	transport (above goal)	3.7	4.4	turn peg	4.6	2.9
retreat peg	3.0–3.2	5.4	3.2	4.5	place (goal)	3.0	4.2	turn peg (unwind)	3.9	4.1

range). Fits with a spectral radius of at least 0.99 or fewer than 500 training steps are left out as not identifiable, which keeps 87 of 98; among those left out is pulling the drawer on the G1 (spectral radius 0.997), so Fig. D.1 says nothing about the drawer contrast of Table D.6. The same Eddie on two robots is at a median descriptor distance of 4.31 (43 pairs, interquartile range 2.40–8.09) and two different Eddies on one robot at 4.91 (831 pairs, 2.95–10.35), not significantly different (Mann-Whitney $p = 0.33$); different Eddies on different robots are at 6.92, which does differ from the first ($p = 0.002$); the MuJoCo and Isaac Sim fits of the same Franka Eddie are at 0.74 (10 pairs). So the same Eddie on two robots is not significantly closer than two different Eddies on one robot (Section 5.3 of the main paper; the test shows only that the shared skill does not make the models alike), and the engine changes the fitted model far less than the robot does.

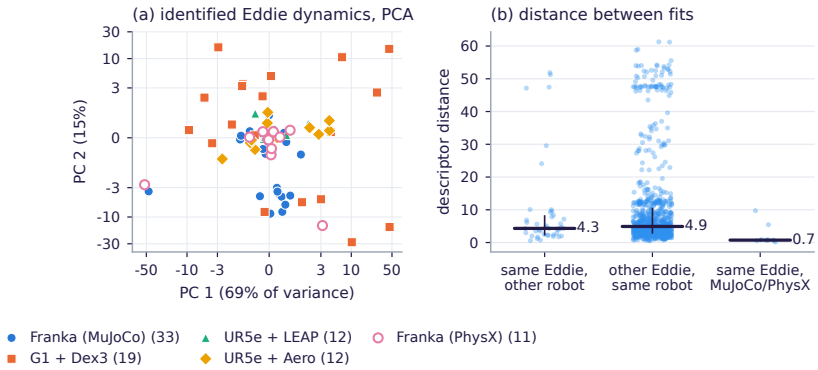


Figure D.1: Dynamics descriptors of the identified Eddies, from their own least-squares fits. (a) PCA of the descriptors, one point per Eddie and robot, on symmetric-log axes so that the G1 outliers do not flatten the rest; Franka Eddies are also shown in Isaac Sim, joined to their MuJoCo point. (b) Distance between the same Eddie on two robots, between two Eddies on one robot, and between the same Eddie in MuJoCo and Isaac Sim (median and interquartile range).

Appendix E. Proofs

The certificates use the models and normalizers of Appendix B and one score per calibration episode $t = 1, \dots, N$, the largest normalized residual over its Eddies and steps,

$$s_t = \max_i \max_k \|W^{1/2} w_k^{(i)}\| / \kappa_i. \tag{E.1}$$

With q the $\lceil (N+1)(1-\alpha) \rceil$-th smallest score, the envelope of Eddie i is $\epsilon_i = q\kappa_i$.

Lemma E.1 (Episode-level coverage) *Let the models (A_i, B_i, d_i) and the normalizers κ_i be fixed. By this we mean that they are fitted on training episodes and calibrated on calibration episodes, that both sets are disjoint from the new episode, and that nothing is changed afterwards. Let the new episode be exchangeable with the calibration episodes. Then*

$$\Pr\left\{ \|W^{1/2} w_k^{(i)}\| \le \epsilon_i \text{ for every Eddie } i \text{ and step } k \right\} \ge 1 - \alpha.$$

Proof of Lemma E.1. With the models and normalizers fixed, the score (E.1) is a fixed function of one episode, so the calibration scores $s_1, \dots, s_N$ and the score s_{N+1} of the new episode are exchangeable. The rank of s_{N+1} among them is uniform on $\{1, \dots, N+1\}$, ties broken at random, so $\Pr\{s_{N+1} \le q\} \ge \lceil (N+1)(1-\alpha) \rceil / (N+1) \ge 1 - \alpha$, with $q = \infty$ if the index exceeds N. Since s_{N+1} is a maximum, $s_{N+1} \le q$ holds if and only if $\|W^{1/2} w_k^{(i)}\| \le q\kappa_i = \epsilon_i$ for every Eddie i and step k. ∎

Let S be a matrix with n columns, either the position map of the main paper or $W^{1/2}$, and let $\sigma_i(X, j) = \sum_{l=0}^{j-1} \|S A_i^l X\|$. The sums

$$g_i(j) = \sigma_i\big(G_i M^{-1/2}, j\big), \qquad \beta_i(j) = \sigma_i\big(B_i M^{-1/2}, j\big), \qquad \omega_i(j) = \sigma_i\big(W^{-1/2}, j\big) \tag{E.2}$$

accumulate over j steps the effect of command changes, of command offsets from the anchor, and of residuals.

Proposition E.2 (Propagated Eddie tubes) *Assume the event of Lemma E.1. Let Eddie i start at step k_0, and let its commands respect the slew bound, $\|e_{k+1} - e_k\|_M \le \rho_i$. Then (4) holds for all $j \ge 0$. Suppose instead that the commands respect the trust radius, $\|e_k - \bar{e}_i\|_M \le \delta_i$. Then the same bound holds for $z_k - z_i^\star(\bar{e}_i)$, with two replacements: $z_{k_0} - z_i^\star(\bar{e}_i)$ takes the place of η_{k_0}, and $\delta_i\beta_i(j)$ takes the place of $\rho_i g_i(j)$.*

Proof of Proposition E.2. By (2), $z_i^\star(e) = A_i z_i^\star(e) + B_i e + d_i$; subtracting this from the model at $e = e_k$ gives $z_{k+1} - z_i^\star(e_k) = A_i\eta_k + w_k$, and since $z_i^\star(e_{k+1}) - z_i^\star(e_k) = G_i(e_{k+1} - e_k)$,

$$\eta_{k+1} = A_i\eta_k - G_i(e_{k+1} - e_k) + w_k,$$

so by induction on j

$$\eta_{k_0+j} = A_i^j\eta_{k_0} + \sum_{l=0}^{j-1} A_i^l\Big(- G_i\,(e_{k_0+j-l} - e_{k_0+j-l-1}) + w_{k_0+j-1-l}\Big).$$

Applying S, the triangle inequality and the operator norm,

$$\|SA_i^j\eta_{k_0}\| \le \|SA_i^jW^{-1/2}\|\,b_{k_0}, \qquad \|SA_i^lG_i\Delta e\| \le \|SA_i^lG_iM^{-1/2}\|\,\rho_i,$$
$$\|SA_i^lw\| \le \|SA_i^lW^{-1/2}\|\,\epsilon_i,$$

where the last bound holds on the event of Lemma E.1; summing gives (4). The trust tube follows in the same way from $z_{k+1} - z_i^\star(\bar{e}_i) = A_i(z_k - z_i^\star(\bar{e}_i)) + B_i(e_k - \bar{e}_i) + w_k$ with $\|e_k - \bar{e}_i\|_M \le \delta_i$. For a stable A_i the transient decays and the sums converge, so the tube settles at the ultimate radius $\rho_i g_i(\infty) + \epsilon_i\omega_i(\infty)$, the radius reported as an Eddie's tube in Tables D.1 and D.6 (the gate evaluates the full bound (4), transient included). ∎

Proof of Theorem 1 of the main paper. Let k be the last step of Eddie i and $k+1$ the first step of Eddie i'. The transition from k to $k+1$ still follows the model of Eddie i, so $z_{k+1} = z_i^\star(e_k) + A_i\eta_k + w_k$ as in the proof of Proposition E.2, and measuring the new error from the equilibrium of Eddie i',

$$\eta_{k+1} = A_i\eta_k + w_k + z_i^\star(e_k) - z_{i'}^\star(e_{k+1}). \tag{E.3}$$

With $a_i = \|W^{1/2}A_iW^{-1/2}\|$ we have $\|W^{1/2}A_i\eta_k\| \le a_ib_k$, and on the event of Lemma E.1 $\|W^{1/2}w_k\| \le \epsilon_i$, so the triangle inequality applied to (E.3) gives (7), which stays valid when b_k is replaced by any upper bound and needs no bound on a_i (Appendix B). Within an Eddie, Proposition E.2 with $S = W^{1/2}$ bounds b at every step by an expression that increases in the starting value, and with the position map S it gives the Eddie's tube. Alternating the two bounds from the measured b at the first step bounds every step of every Eddie, by induction over the Eddies. Every step uses only the residual bound and the issued commands, so the chain holds on the event of Lemma E.1, which has probability at least $1 - \alpha$ under (A1); (A2) makes every equilibrium $z_i^\star$ well defined. An Eddie whose model is not certified (Appendix B) breaks the chain, which restarts from measurement at the next certified Eddie. This proves part (a); parts (b) and (c) are Proposition E.3(a) and (b) below. ∎

Proposition E.3 (Self-calibrating gate) *Consider the gate of Section 4.2 of the main paper, with three rules. It sends e_{k+1} only if (5) holds. It stops when the measured error $\|S\eta_k\|$ exceeds μ. It updates the level by (6) after every episode, with $\mathrm{err}_t = \mathbf{1}\{s_t > q_t\}$. Let the scores satisfy $s_t \in [0, s_{\max}]$, and let $q_1 \in [0, s_{\max}]$. Let ϵ_i be the envelope used in the check, and let it be at least $\max(q_t, 0)\,\kappa_i$; Algorithm 1 widens it to $\max(q_t, s)\,\kappa_i$. Then two claims hold. (a) At every step of an execution that finishes, $\|S\eta_k\| \le \mu$. Moreover, whenever $\|W^{1/2}w_k\| \le \epsilon_i$ for the running Eddie i, the error after the step satisfies $\|S\eta_{k+1}\| \le \mu$. (b) For every T and any sequence of episodes, tasks, models and normalizers, $\left|\frac{1}{T}\sum_{t=1}^{T}\mathrm{err}_t - \alpha\right| \le (s_{\max}+\gamma)/(\gamma T)$.*

Proof of Proposition E.3. (a) The first claim holds because the gate stops when the measured error exceeds μ. For the second, (2) gives $\eta_{k+1} = \hat{\eta}_{k+1} + w_k$ with $\hat{\eta}_{k+1}$ the term from (5), so if $\|W^{1/2}w_k\| \le \epsilon_i$,

$$\|S\eta_{k+1}\| \le \|S\hat{\eta}_{k+1}\| + \|SW^{-1/2}\|\,\|W^{1/2}w_k\| \le \|S\hat{\eta}_{k+1}\| + \epsilon_i\|SW^{-1/2}\| \le \mu,$$

where the last inequality is (5), which holds because the command was sent. (b) First, q_t stays in $[-\gamma\alpha,\ s_{\max}+\gamma(1-\alpha)]$: above $s_{\max}$ no score exceeds q_t, so it falls; below 0 every score exceeds it, so it rises; from inside $[0, s_{\max}]$ one step raises it by $\gamma(1-\alpha)$ or lowers it by $\gamma\alpha$; and $q_1 \in [0, s_{\max}]$. Summing (6) over t gives $q_{T+1} - q_1 = \gamma\sum_t(\mathrm{err}_t - \alpha)$, and since q_{T+1} lies in the interval and q_1 in $[0, s_{\max}]$, $|q_{T+1} - q_1| \le s_{\max}+\gamma$, hence $\left|\frac{1}{T}\sum_t \mathrm{err}_t - \alpha\right| = |q_{T+1}-q_1|/(\gamma T) \le (s_{\max}+\gamma)/(\gamma T)$. This uses nothing about the episodes (Gibbs and Candès, 2021; Angelopoulos et al., 2023). ■

Once the gate refits a model or a normalizer from the task's own episodes, these are no longer fixed in the sense of Lemma E.1, so only Proposition E.3, that is, Theorem 1(b) and (c), still holds. Fig. E.1 shows the level q_t on new tasks, where misses occur at rate α in the long run whatever the prior; at 5 cm the miss rate is 10-12% of episodes (Fig. 7 of the main paper), and from the task's own calibration 8.5-10.5% (Section 5.1).

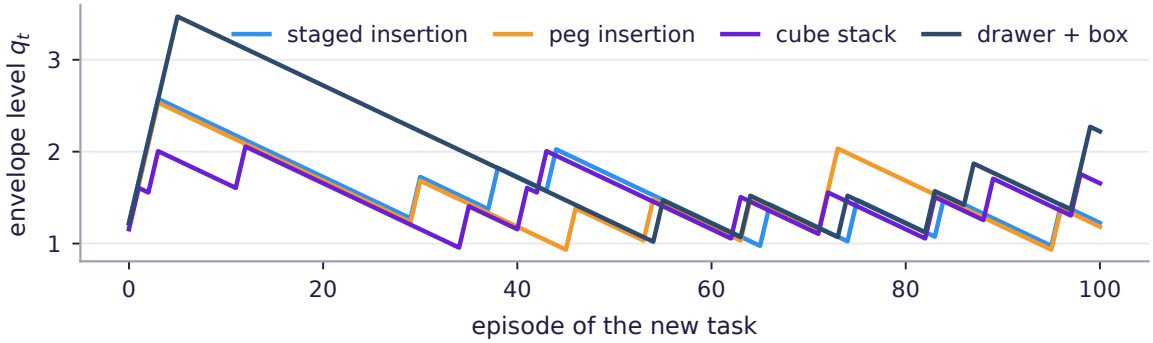


Figure E.1: Coverage level q_t of the self-calibrating gate on each Franka task held out of training ($\mu = 5$ cm), from the level calibrated on the other three tasks; a miss raises q_t by $\gamma(1-\alpha)$ and a covered episode lowers it by $\gamma\alpha$ (6).

Proposition E.4 (Vocabulary growth) *Let every primitive p have a share $\pi_p > 0$ of the error budget, fixed at registration, with $\sum_p \pi_p \le 1$. Let the envelopes of p be calibrated on their own at level $\alpha\pi_p$. This calibration applies (3) to the score $\max_{i\in p}\max_k \|W^{1/2}w_k^{(i)}\|/\kappa_i$, taken over the*

Eddies i of p. Then two claims hold. (a) A new episode stays within every envelope with probability at least $1 - \alpha$. (b) Registering and calibrating a new primitive leaves the envelopes, tubes and chained certificates of all existing primitives unchanged.

Proof of Proposition E.4. (a) Lemma E.1, applied to each primitive on its own, bounds the failure probability of the envelope of p by $\alpha\pi_p$; by the union bound, some envelope fails with probability at most $\alpha \sum_p \pi_p \leq \alpha$. (b) The calibration of p uses only its model, its share and the scores of its own Eddies. A new primitive is accepted only if its share fits in the unallocated budget $1 - \sum_p \pi_p$, so no existing share or envelope changes, and the tubes and chained bounds, which depend only on each primitive's models and envelopes, stay the same too. Section 5.4 of the main paper gives the example of push. ■

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